Module 16: History of Life — Questions

Quick Check

1. Place the following major milestones of life in chronological order from oldest (1) to most recent (4): - ___ First Eukaryotes (cells with nuclei) - ___ First land animals - ___ First Prokaryotes (bacteria) - ___ The Cambrian Explosion

2. The process where a single ancestral species rapidly diversifies into many new species—usually because of a mass extinction event clearing out competition—is called: A) Genetic Drift B) Adaptive Radiation C) Convergent Evolution D) Stabilizing Selection

3. On a phylogenetic tree, the point where a single branch splits into two represents: A) A mass extinction event B) The most recent common ancestor of those two lineages C) An unfavored mutation D) A geographic boundary like a river

4. A scientifically valid "clade" must include: A) Only the currently living species, not the extinct ones. B) An ancestor and ALL of its descendants. C) Animals that look physically similar. D) Animals that live in the same geographic region.

5. True or False: In a phylogenetic tree, spinning the branches around a specific node (flipping the left/right order of the tips) changes the evolutionary relationships. - [ ] True - [ ] False

---

Reflection

6. Approximately 2.7 billion years ago, the "Oxygen Revolution" occurred. What type of organism caused this massive atmospheric shift, and what biological process were they performing to generate oxygen?

{fill:textarea rows=3}

7. Why did it take over 3 billion years for life to finally colonize dry land? (Hint: Think about what the oceans provide, and the danger of the sun's radiation before the ozone layer formed).

{fill:textarea rows=4}

8. If a student looks at the tips of a phylogenetic tree and says "A is closely related to B because they are printed next to each other at the top of the page," why is that student potentially incorrect? Explain the correct way to trace relatedness on a tree.

{fill:textarea rows=5}